IE7275 Machine Learning and Data Analytics

Group Project 1 - Topic Proposal

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Topic: Personalized Hybrid Recommendation System for Amazon Video Games

Dataset

Source: Amazon Reviews 2023 (McAuley Lab, UCSD).

We will use the Video_Games 5-core subset: approximately 94,800 users, 25,600 products, and 814,600 interactions. In a 5-core dataset, each retained user and item has at least five interactions before splitting.

Review interactions (user_id, rating, timestamp) will be joined with product metadata (title, category, features, description) using parent_asin.

Data Preparation

- Inspect sparsity, rating distributions, missing values, and duplicate or invalid interactions.
- Handle missing metadata and prepare the user-item interaction matrix and product text features.
- Use the provided absolute-timestamp chronological train/validation/test split, with training preceding validation and test. Build user histories from training data and tune models on validation data.

Recommendation Models

The focus is Top-K personalized recommendation: rank unseen games a user is likely to interact with next.

- Popularity baseline: recommend the most frequently interacted-with games in the training data.
- Matrix factorization: use implicit ALS to learn user and item preferences from interaction history.
- Neural Collaborative Filtering: learn user-item relationships with a neural network.
- Content-based / hybrid: use TF-IDF on product text for similarity recommendations, then combine content and collaborative scores.

Evaluation

Compare models using Precision@10, Recall@10, and NDCG@10 under the same chronological split and candidate-item protocol, excluding previously seen items. Report ranking quality and briefly discuss computational cost and personalization.

Expected Outcome

We expect the comparison to show whether combining product content with collaborative filtering improves Top-K recommendation relevance, particularly when interaction histories are limited.